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End-functional polymers, thiocarbonylthio group removal/transformation and reversible addition–fragmentation–chain transfer (RAFT) polymerization

Graeme Moad,* Ezio Rizzardo and San H Thang

Abstract

This review focuses on processes for thiocarbonylthio group removal/transformation of polymers synthesized by radical polymerization with reversible addition-fragmentation-chain transfer (RAFT). A variety of processes have now been reported in this context. These include reactions with nucleophiles, radical-induced reactions, thermolysis, electrocyclic reactions and 'click' processes. We also consider the use of RAFT-synthesized polymers in the construction of block or graft copolymers, functional nanoparticles and biopolymer conjugates where transformation of the thiocarbonylthio group is an integral part of the process. This includes the use of RAFT-synthesized polymers in other forms of radical polymerization such as atom transfer radical polymerization or nitroxide-mediated polymerization, and the 'switching' of thiocarbonylthio groups to enable control over polymerization of a wider range of monomers in the RAFT process. With each process we provide information on the scope and, where known, indicate the mechanism, advantages and limitations.
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Keywords: RAFT polymerization; radical polymerization; end-group; polymer modification

INTRODUCTION

Control of radical polymerization by the addition of thiocarbonylthio compounds that serve as reversible addition–fragmentation–chain transfer (RAFT) agents was first reported in 1998.[1,2] The overall RAFT process can be viewed simply as an insertion of monomer units into the C–S bond of a suitable thiocarbonylthio compound, the RAFT agent (**1**), as shown in Scheme 1. A key feature of the process is that the thiocarbonylthio groups, present in the initial RAFT agent (**1**), are retained in the polymer products (**2**). These polymer products being themselves RAFT agents constitute a dormant form of the corresponding propagating radicals. The RAFT process is a reversible deactivation radical polymerization,[3] an extremely versatile process that, with appropriate selection of reagents and reaction conditions, satisfies most of the established criteria for living polymerization.[4,5] It can be applied to form low-dispersity polymers or copolymers from most monomers amenable to radical polymerization.[6–14] The process is compatible with a wide range of functionality in monomers, solvents and initiators such that it is possible to take RAFT polymerizations to high conversion and achieve commercially acceptable polymerization rates. Polymerizations can also be successfully carried out in aqueous or heterogeneous media (emulsion, miniemulsion, suspension). Star, block, microgel and hyperbranched structures, and supramolecular assemblies are accessible and can have high purity.

In some circumstances, it is desirable to transform the thiocarbonylthio group to achieve a desired functionality or for use in subsequent processes post-polymerization. Many examples have now been reported of the use of RAFT-synthesized polymers in the construction of block or graft copolymers, functional nanoparticles and biopolymer conjugates, and many of the more effective strategies require transformation of the thiocarbonylthio end-group as an integral part of the process.

There are also examples where simple removal of the thiocarbonylthio group is desirable. The presence of the thiocarbonylthio groups in RAFT-synthesized polymers means that the polymers may be coloured. The colour may range from violet through red to pale yellow depending on the particular thiocarbonylthio chromophore. The polymers may also, in some cases, release odour over time due to decomposition of the thiocarbonylthio groups and the evolution of volatile sulfur-containing compounds. The thiocarbonylthio group is also known to quench fluorescence[15–18] which is an issue in some optoelectronic applications.[19] While recent studies indicate very low or no cytotoxicity associated with RAFT-synthesized polymers, this issue, nonetheless, needs to be considered particularly in the context of biomedical applications.[20–23] It may also be necessary to deactivate the thiocarbonylthio groups because of their intrinsic reactivity. There are thus a variety of incentives for developing effective methods for treatment of RAFT-made polymers to remove or transform the thiocarbonylthio groups.

The thiocarbonylthio group is not always at the chain end. For example, when symmetrical trithiocarbonates or Z-connected

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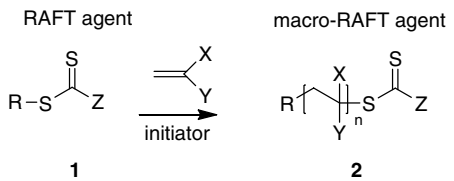
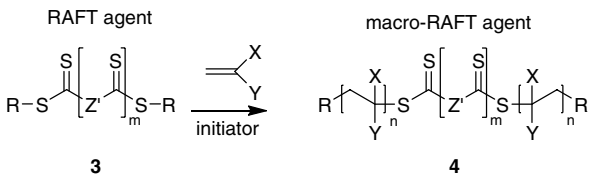
multi-RAFT agents (**3**) are used the functionality will be present in the core and connect the arms of the RAFT-synthesized polymer (**4**; Scheme 2). The chemistries described in this review are also applicable to such polymers. The usual outcome of thiocarbonylthio group removal/transformation applied to such polymers is that the arms are cleaved from the core.

The reactions of the thiocarbonylthio group are well known from small-molecule chemistry[24–27] and much of this knowledge has been shown to be applicable to transforming the thiocarbonylthio groups present in RAFT-synthesized polymers.[1] Many of the methods used for thiocarbonylthio group removal are summarized in Scheme 3. Thiocarbonylthio groups undergo reaction with nucleophiles and ionic reducing agents typically to produce a polymer with a thiol end-group. They also react with various oxidizing agents and are sensitive to UV irradiation. These reactions, which leave reactive end-group functionality, are not appropriate in all circumstances. Thermolysis and radical-induced reactions are other solutions and can provide complete desulfurization.



Scheme 1. Overall RAFT process.



Scheme 2. Overall RAFT process with trithiocarbonate ($m = 0$) or Z-connected bis-RAFT agent ($m = 1$).

The thiocarbonylthio group may also be transformed or, in some cases, be used directly in other forms of radical polymerization, such as atom transfer radical polymerization (ATRP)[28,29] or nitroxide-mediated polymerization (NMP),[30] and, as we have recently demonstrated, certain thiocarbonylthio groups can be 'switched' to enable control over polymerization of a wider range of monomers in the RAFT process.[31,32]

Previous reviews focusing on end-functional polymers and RAFT polymerization include those by Willcock and O'Reilly,[33] Moad *et al.*[34] and Barner and Perrier.[35] Other reviews that include significant sections on end-functional polymers and end-group transformations include our recent reviews of the RAFT process,[6,8–10] that by Boyer *et al.*[13] on biomedical applications and that by Moad *et al.*[19] on optoelectronic applications. Reactions of functionalities attached to the Z or R group of RAFT-synthesized polymers that do not directly affect the thiocarbonyl group are outside the scope of this review. Such processes are covered in other publications.[6,8–10,13,19,33–35]

PROCESSES FOR END-GROUP REMOVAL/TRANSFORMATION

Reaction with nucleophiles

Thiocarbonylthio groups undergo reaction with nucleophiles (e.g. amines,[34,36–49] thiols,[50] hydroxides[43,49–53]) and ionic reducing agents (e.g. boron hydrides[45,51–56]) to provide thiol groups (Scheme 4). A summary of these methods in transforming RAFT-synthesized polymers is provided in Table 1. Hydrolysis is also observed as a side reaction during RAFT polymerization in aqueous media (solution or emulsion polymerization).[41,57–59]

The first applications of nucleophilic thiocarbonylthio group removal from RAFT-synthesized polymers were to characterize the arms of block copolymers formed from symmetrical trithiocarbonates[37] or stars formed from Z-connected multi-RAFT agents.[36] Many have now made use of the method in the synthesis of, for example, gold nanoparticles,[54] biopolymer conjugates[60] and a wide variety of end-functional, block and segmented polymers (Table 1). In these applications end-group fidelity is an important concern. While the initial[36,37] and some more recent studies indicate a clean and quantitative reaction, it is clear that the transformation can be complicated by side reactions. The extent of these depends strongly on the polymer and on the specific reagents and reaction conditions used.[34,40,42,51]

The most well-documented side reaction is oxidative coupling of the initially formed thiol, which is usually manifest by the development of a bimodal molecular weight distribution caused by formation of a disulfide twice the molecular weight of the precursor polymer. The extent of oxidative coupling can be mitigated by conducting the process under an inert atmosphere. Disulfides formed can also be reduced to thiol groups in a subsequent reduction, e.g. with Zn/CH_3CO_2H[34,38] or tris(2-carboxyethyl)phosphine (TCEP).[51] Disulfide formation can be minimized or eliminated by performing aminolysis in the presence

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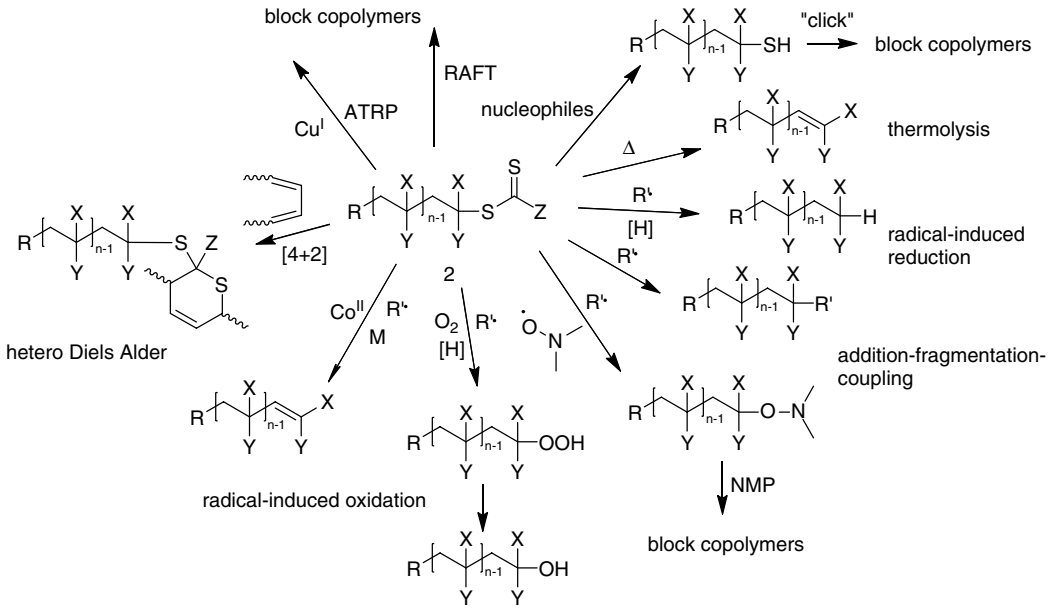
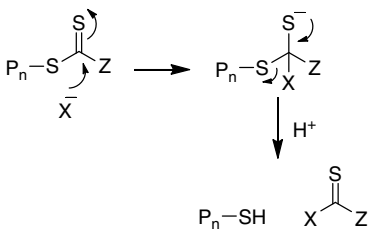


Scheme 3. Processes for RAFT end-group transformation (R′· = radical; [H] = hydrogen donor; M = monomer).



Scheme 4. Reaction of RAFT end-group with nucleophiles.

of reducing agents such as sodium dithionite or sodium bisulfite ($Na_2S_2O_4$, $NaHSO_3$)[40,42,48] or a phosphine (e.g. TCEP).[61,62] It should be pointed out that the sensitivity of the thiol end-group to oxidation particularly in the presence of base means that some care is necessary in handling such polymers.

Two groups have performed a direct comparison of aminolysis with borohydride reduction for polystyrene dithiobenzoate.[63,64] One study[63] compared (a) sodium borohydride in aqueous tetrahydrofuran (THF), (b) sodium borohydride in toluene with phase-transfer catalysis, (c) lithium triethylborohydride in THF and (d) propylamine in THF. The reaction (a) with sodium borohydride in aqueous THF was slow yet yielded the cleanest product. Aminolysis with propylamine (d) gave a substantial yield of the disulfide. The other reductions (b) and (c) gave, in addition, other unidentified by-products. The second study[64] concluded that disulfide formation during borohydride reduction was a problem even in the presence of a phosphine ($P(C_4H_9)_3$).

Very recently it has been reported that the use of hydrazine as a nucleophile is advantageous both in giving rapid rate of reaction and very little disulfide formation even when the process is carried out in air.[65]

The thiol-terminated polymers generated from RAFT-synthesized polymers have been used for a variety of applications. These include their use in thiol-ene processes and in the preparation of biopolymer conjugates. Another example is the oxidation of the thiol-terminated polymers with *m*-chloroperbenzoic acid to provide sulfonic acid-terminated polymers.[66]

Thiol click reactions

Several groups have found that disulfide formation and other side reactions are prevented if the initially formed thiol is allowed to react with a thiol-trapping reagent.[44,64,67–70] The orthogonal nature of the reaction of the thiocarbonylthio compound with nucleophiles and the thiol-trapping reactions allows the two processes to be conducted simultaneously. Reactions used or which could potentially be used in this context are summarized in Scheme 5. They include the thiol-ene reaction,[64,67,68] and reaction with α-bromoesters,[71] isocyanates,[70] methanethiosulfonates[69,72–74] and 2,2′-dithiodipyridine.[68] A summary of these processes is also included in Table 1. The applications of so-called thiol click chemistry in polymer modification have recently been reviewed.[75–77]

Reductive aminolysis in the presence of a Michael acceptor has been applied to the synthesis of three-armed stars by *in situ* thiol-ene reaction with trimethylolpropane triacrylate.[67] A recent paper by Koo *et al.*[78] examined the use of radical-catalysed thiol-ene processes for polymer conjugation. The reaction was found to be problematic because of the incidence of side reactions and difficulties in achieving high conversions unless one reagent was in large excess. The authors concluded that the radical-catalysed thiol-ene reaction should not be considered a 'click' reaction when used for polymer–polymer conjugation.[78]

Aminolysis of poly(*N*-isopropylacrylamide) (PNIPAM) trithiocarbonate in the presence of 2,2′-dithiodipyridine was used to form a polymer with a pyridyl disulfide end-group in very high yield.[68] The thiol end-group can be regenerated under mild conditions and used to make biopolymer conjugates by a disulfide exchange process.

In situ trapping with functional methanethiosulfonates (e.g. **5**, **6**) provides a route to end-functional polymers.[72–74] For example, aminolysis in the presence of *S*-but-3-ynylmethanethiosulfonate (**5**) was used to prepare alkyne end-functional polymers for use in copper-catalysed alkyne-azide cycloaddition 'click' chemistry.[73]

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Table 1. Examples of conversion of thiocarbonylthio end-groups to thiol end-groups by reaction with nucleophiles

Terminal monomer[a]	Z[b]	Reagent(s)[c]	Trapping, etc.[d]	Comment	Ref.
St	Ph	Ethylamine	Zn/CH_3CO_2H	Sequential	38
St	S(star)	Piperidine	–	Arms cleaved	36
St	Ph	Piperidine	–	–	34
St	Ph	$NaBH_4/PBu_3$	Acrylates	*In situ* thiol-ene	64
St	Ph	$NaBH_4/PBu_3$	–	*In situ* thiol-ene	64
St	Ph	$NaBH_4$	–	–	63
St	Ph	$NaBH_4$		–	63
St	Ph	Hexylamine/$NaHSO_3$	–	Subsequently used in disulfide formation	47
St	Ph, SC_2H_5	Hydrazine hydrate		No disulfide formation	65
St/StOAc	Ph	Hydrazine hydrate	–	Simultaneous ester cleavage	186
$StSO_3Na$	Ph	$NaBH_4$		*In situ* reaction with gold surface	53
St/2VP	Ph	Hexylamine/$Na_2S_2O_4$	–	–	187
MMA	Ph	Hexylamine/$Na_2S_2O_4$	–	Subsequently used in thiol-ene	40
MMA	Ph	Hexylamine	–	Used in thiol-ene	79
MMA	Ph	Propylamine	DTP	*In situ* trapping	69
LMA	Ph	Propylamine/$Na_2S_2O_4$	–	–	48
DMAEMA	Ph	Hexylamine/$NaHSO_3$	–	Subsequently used in disulfide formation	47
MAA	Ph	NaOH		[e]	22
HPMA	Ph	$NaBH_4$		[e]	22
MA	S(star)	Piperidine	–	Arms cleaved	36
MA	S(PMA)	Ethylenediamine	–	Arms cleaved	37
MA	Ph	Hydrazine hydrate		No disulfide formation	65
MA	Ph	Hexylamine	Et_3N/MBP, Et_3N/EBiB	Thiobromo 'click'	71
BA	SC_4H_9	KOH	–	–	188
AA	Ph	Hexylamine/$NaHSO_3$	–	Subsequently used in disulfide formation	47
PEGA	Ph	Butylamine		Subsequently used in thiol-ene	21
NIPAM	$SCH_2CH(CH_3)_2$	Butylamine/TCEP	HEA	One-pot thiol-ene	44
NIPAM	$SCH_2CH(CH_3)_2$	Butylamine/TCEP	BA	One-pot thiol-ene	44
NIPAM	$SCH_2CH(CH_3)_2$	Butylamine/TCEP	–	Used in thiol-ene	61, 62
NIPAM	$S(CH_2)_2CO_2H$	2-Aminoethanol	DTP	*In situ* trapping	68
NIPAM	Pyrrole	NaOH/EDTA	–	–	189
NIPAM	Ph	$NaBH_4$	–	Gold nanoparticles	54
NIPAM	Ph	$LiB(C_2H_5)_3H$	–	Subsequently used in thiol-ene	56
NIPAM	$S(CH_2)_2CO_2H$	$NaBH_4$	–	Subsequently used in thiol-ene	55
NIPAM	SC_4H_9	NaOH/MeOH/EDTA	–	Subsequently used in thiol-ene	190
NIPAM	Ph	Octylamine/DMPP	AMA	*In situ* thiol-ene	49
NIPAM	Ph	Octylamine/DMPP	PA	*In situ* thiol-ene	49
NIPAM	Ph	Hexylamine	Et_3N/MBP, Et_3N/EBiB	Thiobromo 'click'	71
NIPAM	SC_2H_5	Hydrazine hydrate		No disulfide formation	65
DEAM	Ph	Hexylamine/DMPP	TMPTA	*In situ* thiol-ene	67
DMAM	Ph	$NaBH_4$		*In situ* reaction with gold surface	53, 191
AMPS	Ph	NH_3	–	Kinetic study	41
AN	Ph	Propylamine	MTS		74
NVP	OC_2H_5	$NaBH_4$		Used in peptide/nucleotide conjugation	60

[a] Monomer unit adjacent to thiocarbonylthio group.
[b] Z group of RAFT agent **1** or macro-RAFT agent **2**.
[c] Nucleophile/reducing agent: ethylenediaminetetraacetic acid (EDTA), TCEP, dimethylphenylphosphine (DMPP).
[d] Reagent used to trap initially formed thiol: 2,2′-dithiodipyridine (DTP), ethyl 2-bromoisobutyrate (EBiB), trimethylolpropane triacrylate (TMPTA), methyl 2-bromopropionate (MBP), methyl methanethiosulfonate (MTS).
[e] During conversion of RAFT-synthesized poly(pentafluorophenyl methacrylate) to poly(methacrylic acid) (by reaction with NaOH) or a polymethacrylamide derivative (by reaction with a range of primary amines, such as 2-hydroxypropylamine). Conversion efficiency to thiol end-group was determined as >80%.
Other abbreviations not given elsewhere: StOAc, 4-acetoxystyrene; 2VP, 2-vinylpyridine; DMAEMA, 2-(dimethylamino)ethyl methacrylate; HPMA, hydroxypropyl methacrylate; AA, acrylic acid; NIPAM, *N*-isopropylacrylamide; PEGA, poly(ethylene glycol) monomethyl ether acrylate; HEA, 2-hydroxyethyl acrylate; AMA, allyl methacrylate; PA, propargyl acrylate; DEAM, *N*,*N*-diethylacrylamide; DMAM, *N*,*N*-dimethylacrylamide; AMPS, sodium 2-acrylamido-2-methylpropanesulfonate.

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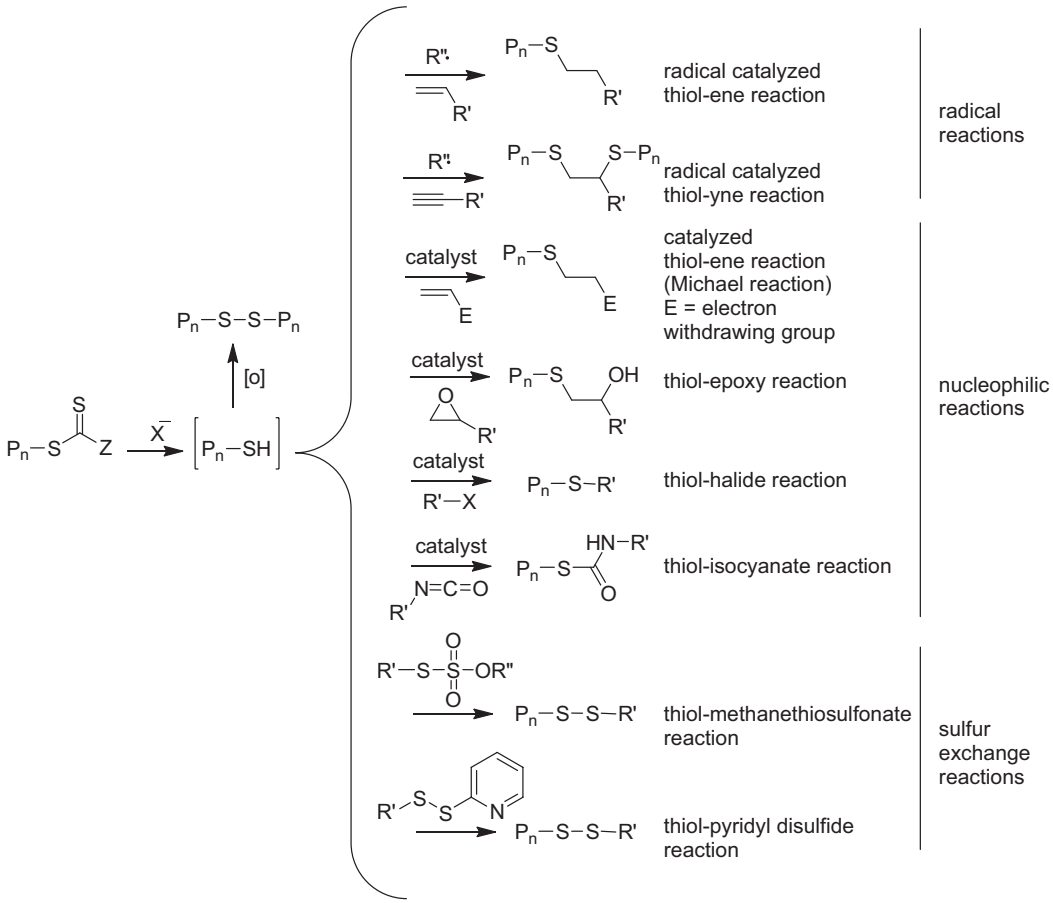
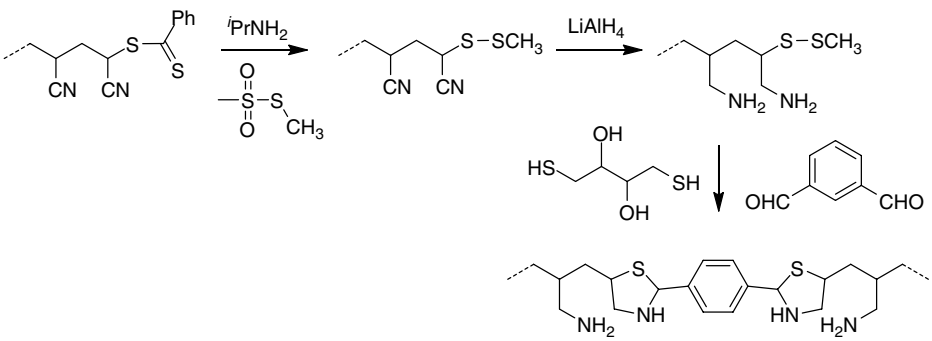


Scheme 5. Reaction of RAFT end-group with nucleophiles with trapping of thiol end-group.



Scheme 6. Coupling polymers with terminal acrylonitrile units using thiazolidine chemistry.

The thiol end-group can also be regenerated from the methyldisulfide end-group by treating the polymer with dithiothreitol ((2*S*,3*S*)-1,4-bis-sulfanylbutane-2,3-diol).[74] This was used as the basis for further end-group transformation using thiazolidine chemistry (Scheme 6).[74]

A RAFT-based approach to multiblock copolymers is based on purposely carrying out oxidative coupling of polymers with thiol end-groups (Scheme 7).[47,79–84] The telechelics are synthesized by RAFT polymerization making use of a bis-RAFT agent with subsequent end-group transformation to form the bis-thiol. The procedure was also applied to synthesize cyclic polymers under high dilution conditions.[80]

Other side reactions

A recent study proposed that dithioester and trithiocarbonate RAFT agents (R–SC(S)Z) can undergo a radical-catalysed reaction with thiols which results in oxidation of the thiol to a disulfide and concomitant reduction of the RAFT agent to R–H (Scheme 8).[50] The relative rates of reaction suggest that dithioester RAFT agents should be preferred and symmetrical trithiocarbonates avoided when preparing polymers with terminal thiol functionality by RAFT polymerization followed by nucleophilic cleavage of the thiocarbonylthio group. To minimize the likelihood of this process, it is important that the RAFT-synthesized polymer is free of any residual initiator.

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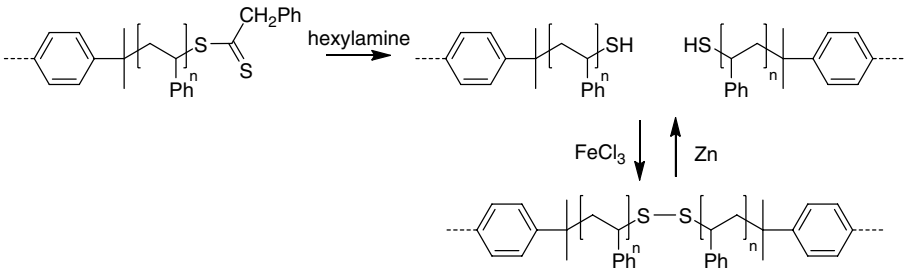
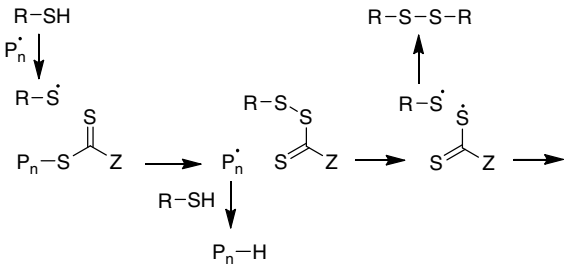
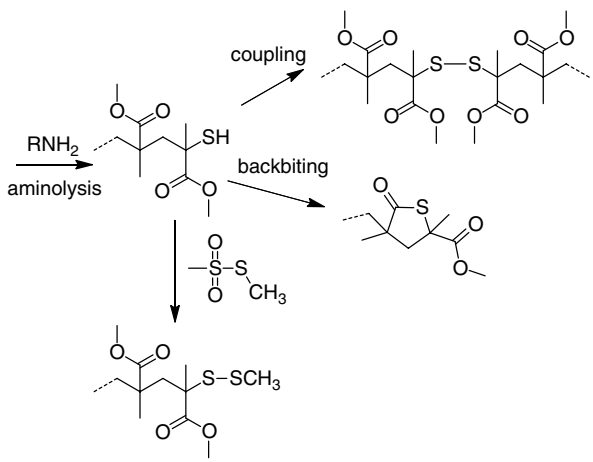


Scheme 7. Reversible coupling of polymers with thiol end-groups.[80]



Scheme 8. Radical-induced reaction of RAFT agents with thiols.



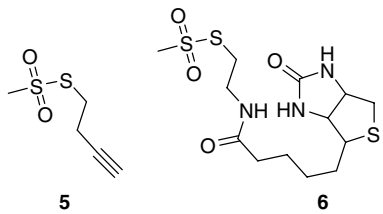
Scheme 9. PMMA RAFT end-group transformation.

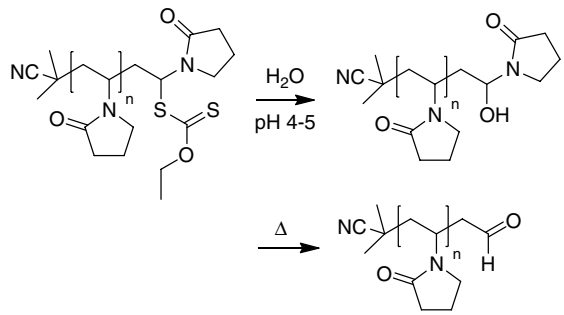
Aminolysis of poly(methyl methacrylate) (PMMA) dithiobenzoate can be further complicated by the thiol undergoing a backbiting reaction to give a thiolactone chain end.[46,69] This side reaction can be avoided by trapping the initially formed thiol. Aminolysis in the presence of *S*-methylmethanethiosulfonate was reported to quantitatively produce PMMA with methanedithio chain ends even in non-degassed media (Scheme 9).[69]

Aminolysis of RAFT-synthesized polymers produces, in addition to the desired polymer, a dithiocarbamate which can sometimes be difficult to separate. The issue of removal of by-products following end-group removal can be resolved through use of a functional reagent (e.g. a functional amine such as 2-(*N*, *N*-dimethylamino)ethylamine)[85] or a RAFT agent with appropriate functionality on Z.[86]

N-vinylpyrrolidone (NVP)

The chain ends of poly(*N*-vinylpyrrolidone) (PNVP) formed with xanthate RAFT agents are susceptible to hydrolysis and can





Scheme 10. Poly(vinylpyrrolidone) end-group hydrolysis.

be converted to aldehyde chain ends (Scheme 10) under mild conditions (40 °C in water).[87] Pound *et al.*[87] took advantage of this in preparing a PNVP with aldehyde chain ends which was then used to form a conjugate with lysozyme hydrochloride by reductive amination.

Xanthate end-group stability also appears to be an issue during RAFT polymerization of NVP and similar polymers particularly at higher reaction temperatures (>60 °C) and/or in aqueous media.[88] Nonetheless, PNVP with high end-group fidelity can be successfully prepared by RAFT polymerization at 60 °C.[60,89]. Zelikin *et al.*[60] cleaved the xanthate end-groups of PNVP by reduction with sodium borohydride and immediately protected the thiol end-groups formed by reaction with Ellman's reagent. The thiol end-groups were subsequently liberated by treatment with DTT and used in bioconjugation.

Reaction with oxidizing agents

RAFT agents and thiocarbonylthio end-groups of macro-RAFT agents react with various oxidizing agents to modify or cleave thiocarbonylthio functionality. These reagents include ozone,[90,91] air,[92–95] hydrogen peroxide,[96] hydroperoxides[94,95,97] and peracids.[98,99] They are also sensitive to UV irradiation.[100,101] Examples of these processes are provided in Table 2.

RAFT-synthesized polymers with xanthate end-groups react with ozone quantitatively to form the corresponding poly-

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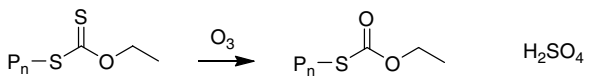
Table 2. Examples of oxidation of thiocarbonylthio end-groups

Terminal monomer[a]	Z[b]	Reagent	Comment	Product end-group	Ref.
StMe	OEt	Ozone	–	-SC(O)OEt	90
BA	OEt	Ozone	–	-SC(O)OEt	91
MA	Ph	*t*-BuOOH	Sulfine intermediate	-SC(O)Ph	92
iBoA	2-Pyridyl	Air	AIBN, THF solvent	—OOH	94
MMA, acrylates	Ph	Air	AIBN, THF solvent	—OOH	94, 95
St, acrylates	SP_n	Air	AIBN, THF solvent	—OOH	95
MA, BA	CH_2Ph	Air	AIBN, THF solvent	—OOH	95
S	Ph	H_2O_2	–	—OH	96
NVP	OEt	H_2O_2	–	—OH	96

[a] Monomer unit adjacent to thiocarbonylthio group.
[b] Z group of RAFT agent **1** or macro-RAFT agent **2**.
Abbreviations not defined elsewhere: StMe, 4-methylstyrene; iBoA, isobornyl acrylate.



Scheme 11. Reaction of xanthates with ozone.

mer with a thiolocarbonate end-group.[90,91] The mechanism has not been fully elucidated but the thiocarbonyl is replaced with a carbonyl and sulfuric acid is formed as a by-product as indicated in Scheme 11. The process was claimed to also be applicable to other RAFT end-groups (trithiocarbonates, dithioesters) and a patent was filed on the use of this process.[91]

Thiocarbonylthio chain ends of RAFT-synthesized polymers can be efficiently converted to hydroxyl chain ends in high yield by heating a THF solution of the polymer and azobis(isobutyronitrile) (AIBN) in air (to produce a polymer with hydroperoxy ends) then treating this solution with triphenylphosphine.[94,95] The process was demonstrated for polymers with dithiobenzoate or dithiophenylacetate end-groups and for polymers with trithiocarbonate mid groups (synthesized with dibenzyltrithiocarbonate) (Table 2).

Dithioester end-groups have been shown to react directly with hydroperoxides (in particular, *t*-butylhydroperoxide) to produce a sulfine end-group in high yield.[92] This transformation of dithioester or trithiocarbonate end-groups to sulfine end-groups has precedent in the similar transformation of low-molecular-weight thiocarbonylthio compounds.[92,98,99]

It has also been proposed that hydrogen peroxide can be used to directly transform RAFT end-groups to hydroxyl end-groups.[96] The process was applied to PNVP with xanthate chain ends and polystyrene (PSt) with dithiobenzoate end-groups and involved heating the polymers with hydrogen peroxide at 60 °C. The proposed mechanism involved thermal generation of hydroxyl radicals and an addition–fragmentation–coupling process (see below). However, while hydrogen peroxide may undergo induced decomposition, thermal homolysis requires relatively high temperatures.

RAFT-synthesized polymers, in particular PSt or PMMA with dithiobenzoate chain ends, have been found to be unstable to storage in cyclic ether solvents (THF).[97] Thus PMMA dithiobenzoate was converted to PMMA hydroperoxide within 7 days. PSt dithiobenzoate was more slowly converted to a mixture of products in which the dithiobenzoate end-group had been converted to the corresponding sulfine or thionoester end-groups or was lost and the polymer left with epoxy and unsaturated end-groups. The end-group degradation was attributed to the presence of THF hydroperoxide in the solution. Much slower degradation was also observed in dichloromethane solution. The result has implications with respect to storage of samples for gel permeation chromatography where THF is commonly used as solvent/eluent. We also have observed that RAFT-synthesized polymers (not just dithiobenzoates) can be unstable in THF solution and recommend that only peroxide-free solvent is used and that samples, if they must be kept, should be stored under an inert atmosphere and protected from light.

Table 3. Transfer constants (C_{tr}) and rate constants for transfer to solvent (k_{tr}) in radical polymerization at 80 °C[a]

	Monomer					
	Ethyl acrylate		MMA		St	
Solvent	C_{tr}[b] ($\times 10^4$)	k_{tr}[c] (L mol^{-1} s^{-1})	C_{tr}[b] ($\times 10^4$)	k_{tr}[c] (L mol^{-1} s^{-1})	C_{tr}[b] ($\times 10^4$)	k_{tr}[c] (L mol^{-1} s^{-1})
Toluene	2.6	11.3	0.3–0.9	0.078	0.2–0.8	0.033
Isopropanol	28.7	125	1.9	0.25	4	0.27

[a] Table reproduced from Chong *et al.*[102] © American Chemical Society.
[b] The transfer constant (C_{tr}) is the ratio of the rate constant for hydrogen transfer from the solvent (k_{tr}) to that for propagation in radical polymerization (k_p). Values are taken from the Polymer Handbook.[192]
[c] Values of k_{tr} calculated from the median value of C_{tr} shown using values of k_p for bulk polymerization at 80 °C.[193]

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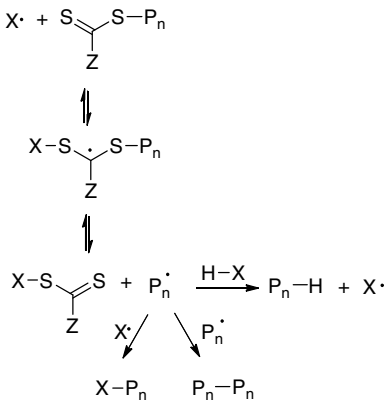

Scheme 12. Radical-induced reduction of thiocarbonylthio compounds.

Radical-induced reduction

Radical-induced reduction allows the thiocarbonylthio group of RAFT-synthesized polymers to be replaced with a hydrogen atom.[16,102–104] The process of radical-induced reduction makes use of the property of thiocarbonylthio compounds to undergo facile reaction with radicals by addition–fragmentation. A general mechanism for the process is shown in Scheme 12. The radical (X•) adds to the thiocarbonyl group of the macro-RAFT agent to generate an intermediate which fragments to provide a propagating radical and a new thiocarbonylthio compound. Reaction of the propagating radical with a hydrogen atom donor (H–X) then gives the desired product in which the thiocarbonylthio group has been replaced with hydrogen. The reaction is a radical chain reaction and is catalytic in X•. Not shown in the scheme is the initiation step for generation of radicals. In principle, non-hydrogen-donor transfer agents can also be used in the process to produce end-functional polymers.

Radical-induced reduction of low-molecular-weight thiocarbonylthio compounds is well known.[105] Radical-induced reduction of xanthates is the basis of the Barton–McCombie reaction for deoxygenation of secondary alcohols (where the xanthate *O*-alkyl is a good homolytic leaving group).[106,107]

Various side reactions can complicate the process; these include the various radical–radical termination reactions. Shown in Scheme 12 are reactions of the intermediate species or the propagating radical with other radical species (initiator-derived radicals, transfer agent-derived radicals, propagating radicals). The reactions are shown to proceed by combination. However, reaction by disproportionation is also likely, the extent being dependent on the particular species involved.

Termination reactions will be favoured by a high radical flux which is usually associated with a high initiator concentration. Attempted radical-induced reduction of poly(butyl acrylate) (PBA) with trithiocarbonate ends in toluene as a hydrogen-donor solvent and with dibenzoyl peroxide (BPO) or didodecyl peroxide (LPO) as initiator (two- to threefold molar excess over end-groups) gave complete end-group removal but a product with a distinctly bimodal molecular weight distribution.[102] The formation of a polymer with peak molecular weight approximately twice that of the precursor polymer was indicative of ineffective radical trapping. NMR analysis showed, in addition to the desired–H end-groups, end-groups formed by transfer to initiator. The reduction was more effective with toluene/isopropanol (1 : 1) as solvent though broadening of the molecular weight distribution was still evident. This is in accord with isopropanol being *ca* 10-fold more effective as a hydrogen donor towards the PBA propagating radical. The transfer constants in radical polymerization ($C_{tr} = k_{tr}/k_p$) of toluene and isopropanol provide an indication of the relative ability of toluene and isopropanol to act as hydrogen donors towards propagating radicals (Table 3). Radical-induced reduction under these conditions, while effective in removing the thiocarbonylthio end-groups, is not recommended as a means of producing polymers with–H end-groups.

Radical-induced reduction is more effective with more active hydrogen-donor transfer agents such as stannanes, silanes and hypophosphites (Table 4).[102] The end-groups of the (meth)acrylic polymers, e.g. PBA and PMMA, are more readily reduced than those of PSt and the efficiency of hydrogen atom donors increases in the series toluene < 2-propanol < triethylsilane < triphenylsilane ≈ tris(trimethylsilyl)silane ≈ *N*-ethylpiperidine hypophosphite < tri-*n*-butylstannane.[102] Tin hydrides, while very effective hydrogen donors, have issues with toxicity and by-product removal and a variety of tin hydride substitutes have been reported.[105] Hypophosphite salts, in particular *N*-ethylpiperidine hypophosphite,[108] are also very effective hydrogen donors and, for organic-soluble polymers, work-up is simplified to a water wash.

Peroxide initiators (LPO, BPO) appear generally more effective as initiators in radical-induced reduction than azo compounds

Table 4. Examples of end-group removal by radical-induced reduction

Terminal monomer[a]	Z[b]	Initiator	H-donor	[H-donor]/[RAFT]	Temperature (°C)	Ends removed (%)	Ref.
BA	$SC_{12}H_{25}$	AIBN	TTMSS	1.5	60	100	102
St	$SC_{12}H_{25}$	AIBN	TTMSS	2	80	100	102
St	SC_4H_9	AIBN	TTMSS	5	70	*ca* 80	143
St	SC_4H_9	AIBN	Bu_3SnH	5	70	100	143
*t*BA	$SC_{12}H_{25}$	AIBN	EPHP	5	100	100	194
BA	$SC_{12}H_{25}$	ACHN	EPHP	5	100	100	102
MMA	$SC_{12}H_{25}$	ACHN	EPHP	5	100	100	102
St	$SC_{12}H_{25}$	ACHN	EPHP	10	100	100	102
St	Ph	ACHN	EPHP	25	110	100	102

[a] Monomer unit adjacent to thiocarbonylthio group.
[b] Z group of RAFT agent **1** or macro-RAFT agent **2**.
Abbreviations not defined elsewhere: EPHP, *N*-ethylpiperidine hypophosphite; TTMSS, tris(trimethylsilyl)silane; ACHN, 1,1′-azobis(1-cyclohexanecarbonitrile).

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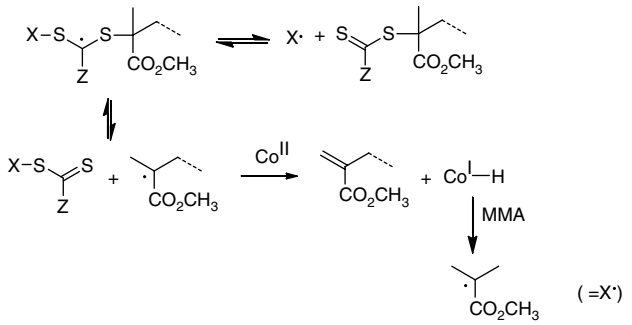


Scheme 13. RAFT catalytic chain transfer process.

such as AIBN.[102] This is attributed to the peroxide-derived radicals being more effective in displacing the propagating radicals than the cyanoalkyl radical (see the section on addition–fragmentation–coupling below). However, the peroxide initiators are also prone to provide end-groups by transfer to initiator.

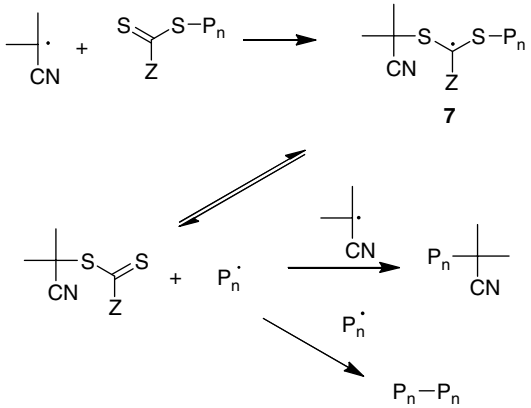
RAFT agents and catalytic chain transfer

Catalytic chain transfer (CCT) polymerization is an important polymerization technique, introduced in the mid-1970s to control the molecular weight and end-group functionality of polymers.[109] CCT is effective in the polymerization and copolymerization of methacrylic and styrenic monomers and results in the formation of polymers (macromonomers) with very high end-group fidelity.[109,110] The CCT agents used in the process are most often square planar Co^{II} or Co^{III} complexes. The macromonomers can be used in the synthesis of low-dispersity block copolymers using the macromonomer RAFT process,[111,112] as chain transfer agents[113,114] and in the preparation of graft copolymers.[115]

Soeriyadi *et al.*[116] have reported that addition of a very small amount of a CCT agent ([$(CH_3OH)_2$Co(dimethyl glyoxime $BF_2)_2$]:[RAFT agent]$_0$ = 1:80 000) during the latter stages (*ca* 70% monomer conversion) of a RAFT polymerization of a methacrylate monomer (methyl methacrylate (MMA) or butyl methacrylate) provides complete end-group removal and a high-purity low-dispersity macromonomer product. Details of the mechanism were not presented. A possible mechanism which combines elements of those for the RAFT and CCT processes is shown in Scheme 13. The process could be considered as an example of radical-induced oxidation.

Radical addition–fragmentation–coupling

One convenient method of RAFT agent synthesis is radical-induced ester exchange in which a RAFT agent is heated with an excess of a radical initiator (e.g. AIBN; Scheme 14).[117–119] For example, the process was used to prepare 2-cyanoprop-2-yl dithiobenzoate from cumyl dithiobenzoate in >90% isolated yield with AIBN as a source of free radicals.[117] Perrier *et al.*[120,121] have reported that an analogous process can be used to remove thiocarbonylthio end-groups from a wide range of RAFT-synthesized polymers (PMMA, PSt, poly(methyl acrylate) (PMA)) with recovery of a RAFT agent. This method is now one of the more widely used methods for transforming thiocarbonylthio groups from RAFT-synthesized polymers (Table 5).[120,122–147] The process typically involves heating (60–90 °C) a solution of the polymer with an azo initiator (most often AIBN) in very large excess (≥20:1 molar ratio



Scheme 14. Addition–fragmentation–coupling (AIBN initiator).

of initiator to polymer). The process is reported to be effective with dithiobenzoates and trithiocarbonates. The process was not effective for removing xanthate chain ends from poly(vinyl acetate) (PVAc).[104] In this case end-group removal most likely fails because the cyanoisopropyl radical (a very good radical leaving group) is not effective in displacing the PVAc propagating radical.

We have found in our own work that heating a polymer with a large excess of AIBN or another azo compound, while successful for removing the chain ends of polymers with a terminal methacrylate unit, gave incomplete removal of dithiobenzoate or butyl trithiocarbonate ends from polymers with a terminal butyl acrylate (BA)[122] or styrene (St) unit;[122,143] even when a very large molar excess, up to 100-fold, of initiator (AIBN) and extended reaction times were used. The relative leaving group ability of $P_n^{\bullet}$ and $X^{\bullet}$ is crucial in determining the outcome of radical addition–fragmentation–coupling. Selection of the radical initiator(s) thus determines the effectiveness of removal of the thiocarbonylthio group. Incomplete thiocarbonylthio group removal from polystyrenes and polyacrylates[122,143] is attributed to the cyanoisopropyl radical being a much better homolytic leaving group than the corresponding radicals ($P_n^{\bullet}$) and the intermediate **7** partitioning in favour of the cyanoisopropyl radical.

Use of LPO[102,122] or BPO[102,123] as initiator completely removes the thiocarbonylthio group. The undecyl radical (from LPO) or phenyl and benzoyloxy radicals (from BPO) are poor radical leaving groups and their addition to the thiocarbonylthio group is irreversible. However, a pronounced bimodal molecular weight distribution due the prevalence of $P_n^{\bullet}$ coupling may be seen when using these initiators (see the section on radical-induced reduction). A combination of LPO (2 molar equivalents) and AIBN (20 molar equivalents) was found to completely remove the thiocarbonylthio end-groups of PSt and polyacrylates with minimal formation of bimodal molecular weight distributions (Scheme 15).[122]

A cautionary note when using this method for introducing functionality by end-group transformation is that the reaction of (for example) cyanoisopropyl radicals with propagating radicals may involve disproportionation as well as the intended combination process. For example, the value of the combination/disproportionation ratio (k_{tc}:k_{td}) for cyanoisopropyl reacting with PSt propagating radical is 0.61.[148] This is an important consideration in using the method in the synthesis of end-functional polymers when high end-group fidelity is required.

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Table 5. Examples of end-group removal by addition–fragmentation–coupling

Terminal monomer[a]	Z[b]	Initiator	[Initiator]/[RAFT][c]	Temperature (°C)	Ends removed[d] (%)	Ref.
MA	Ph	AIBN	20	80	(100)	120
MA	Ph	ACHN	20	80	(100)	120
MA	Ph	ACPA	20	80	(100)	120
BA	Ph	AIBN			[e]	122
HA	Ph	ACPA	5	70	100	124
HA	Ph	AHPN	5	70	100	124
BA/NA	Ph	AIBN	20	80	100	125
PEGA/DMAEA	Ph	AIBN	20	80	(100)	120
PEGA	Ph	ACHN	20	80	(100)	120
BA	SC_4H_9	AIBN	20–200	80	[e]	122
BA	SC_4H_9	AIBN+BPO	20 + 2[f]	80	100	122
HEA	$SC_{12}H_{25}$	AIBN	20	70	100	126
HPA	$SC_{12}H_{25}$	AIBN	10	80	(100)[g]	123
HPA	$SC_{12}H_{25}$	BPO	10	80	100	123
2EHA	Ph	T21S	20	80	100	127
*t*BA/TBBPMA	Ph	AIBN	(20)	(80)	100	128
DAMEA/TFEA	$SC_{12}H_{25}$	AIBN	20	65	100	129
HADA/MADA/NLAA	$SC_{12}H_{25}$	AIBN	20	60	100	130
MMA	Ph	AIBN	20	80	100	120, 131
MMA	Ph	**9**	20	80	100	131
OEGMA	Ph	AIBN	30	80	100	132
OEGMA	Ph	**9**	30	80	100	132
MMA	$C_{10}H_7$	AIBN[h]	20	65	100	133
LMA	Ph	AIBN	30	80	100	134
HADM/MADM/NLAM	Ph	AIBN	20	60	100	130
8	Ph	ACPA	34	80	100	144, 145
NIPAM	SC_2H_5	**10**	21	70	100	146, 147
NIPAM	$SC_{12}H_{25}$	AIBN	(20)	(80)	(100)	135
NIPAM	S(PDMAM-*b*-PNIPAM)	AIBN	(20)	80	(100)	136
NAM/NAS	$SC_{12}H_{25}$	AIBN	400	80	(100)[i]	137
DMAPMAM	Ph	AIBN	20	70	100	138
HPMAM/NMS	Ph	AIBN	20	80	100	139
HPMAM/APMAM	Ph	AIBN	20	70	100	140
DMAM	S(PDMAM)	AIBN	(20)	80	(100)	136
St	Ph	AIBN	28	90	100	120, 142
St	Ph	AIBN	20–100	80	[e]	122
St	Ph	AIBN+BPO	20 + 2[f]	80	100	122
St	Ph	**11**	25	100	100	141
St	SCH_3	AIBN	20	80	(100)	120
St	SC_4H_9	AIBN			[e]	122
St	SC_4H_9	AIBN	20	80	*ca* 95[j]	143
St	S(PSt)	AIBN	(20)	80	(100)	136
VAc	C_2H_5O	AIBN	–	–	[k]	104

[a] Monomer unit adjacent to thiocarbonylthio group.
[b] Z group of RAFT agent **1** or macro-RAFT agent **2**.
[c] Mole ratio of initiator to RAFT-synthesized polymer.
[d] Percentage of end-groups removed. Where details of polymer characterization were not provided, the values appear in parentheses.
[e] Incomplete end-group removal.
[f] A mixture of AIBN and BPO was used.
[g] Bimodal distribution observed for higher reaction temperatures (>80 °C).
[h] In supercritical CO_2 (310 bar (31 MPa)).
[i] Complete end-group removal after repeated reaction.
[j] The amount of AIBN used is incorrectly reported in Postma *et al.*[143]
[k] End-group not removed.
Abbreviations not defined elsewhere: ACPA, 4,4′-azobis(4-cyanopentanoic acid); AHPN, 2,2′-azobis(5-hydroxy-2-methylpentanenitrile); NA, 9-acryloyloxy-[3,3-bis(4-methoxyphenyl)]-3*H*-naphtho[2,1-*b*]pyran; DMAEA, 2-dimethylaminoethyl acrylate; T21S, *tert-butyl 2-ethylhexaneperoxoate*; TBBPMA, *tert*-butyl 2-(((2-bromopropanoyl)oxy)methyl)acrylate; TFEA, trifluoroethyl acrylate; HADA, 3-hydroxyadamantan-1-yl acrylate; MADA, 2-methyladamantan-2-yl acrylate; MADM, 2-methyladamantan-2-yl methacrylate; 5-acryloyloxy-2,6-norbornanelactone; OEGMA, oligo(ethylene glycol) methacrylate; LMA, lauryl methacrylate; HADM, 3-hydroxyadamantan-1-yl methacrylate; 5-methacryloyloxy-2,6-norbornanelactone; poly(*N*,*N*-dimethylacrylamide); DMAPMAM, *N*-(3-(dimethylamino)propyl)methacrylamide hydrochloride; HPMAM, *N*-(2-hydroxypropyl)methacrylamide; APMAM, *N*-(3-aminopropyl)methacrylamide hydrochloride; NAS, *N*-acryloyloxysuccinimide; NMS, *N*-methacryloyloxysuccinimide; HPA, hydroxylpropyl acrylate.

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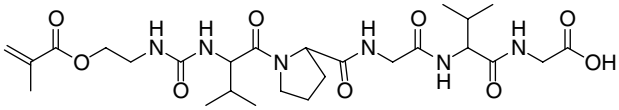
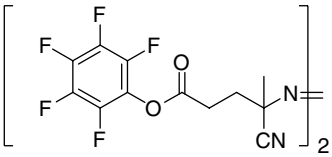


8 peptide macromonomer[144,145]



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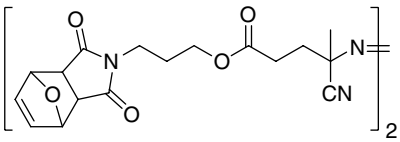

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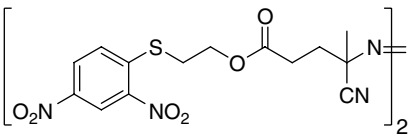

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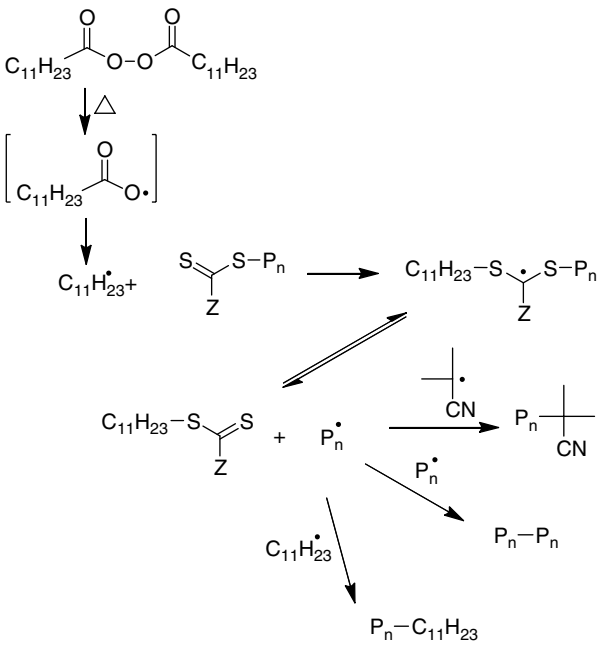
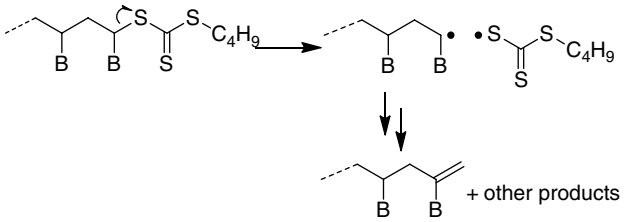


Scheme 15. Addition–fragmentation–coupling (AIBN + LPO initiator).

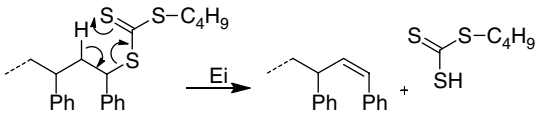
Thermolysis

Thermolysis potentially provides a method for complete thiocarbonylthio group removal.[34,89,143,149,150] However, the dominant mechanism of end-group loss depends on both the particular polymer and the type of thiocarbonylthio group.

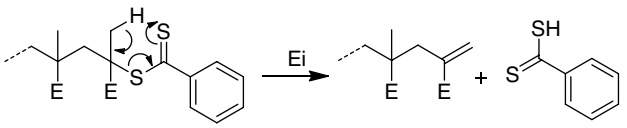
Butyl trithiocarbonate groups are cleaved from RAFT-synthesized PBA[89,149] by heating at *ca* 180 °C in the absence of air. The product is a macromonomer with an exo-methylene double bond formed by consecutive C–S bond homolysis, backbiting and β-scission (Scheme 16). This mechanism is analogous to that proposed[151,152] for thermal degradation of PBA and other acrylate esters made by conventional radical polymerization at higher temperatures (304–370 °C). End-group purities were between 65 and 90%.[89] Thermolysis of RAFT-synthesized PBA may provide a route to low-dispersity acrylic macromonomers.



Scheme 16. Thermolysis of PBA with *S*-butyl trithiocarbonate end (B = $CO_2C_4H_9$).



Scheme 17. Thermolysis of polystyrene with *S*-butyl trithiocarbonate end.



Scheme 18. Thermolysis of PMMA with dithiobenzoate end (E = CO_2CH_3).

Isothermal thermolysis of PSt with butyl trithiocarbonate ends at temperatures in the range 210–250 °C under nitrogen cleanly eliminating the butyl trithiocarbonate group from the yellow PSt to provide a colourless product.[34,143,149] The mechanism appears to be a concerted Chugaev elimination process (Scheme 17).[153,154] In the presence of copper powder loss of the trithiocarbonate group occurs at a significantly lower temperature (onset temperature of 165 °C) and the thermolysis mechanism is modified such that the major product has an exo-methylene double bond (identified by ^{1}H NMR).[143] It was proposed[143] that the copper-catalysed process involves homolysis of the C–S bond to give a PSt propagating radical which can then decay by backbiting and chain scission analogous to that shown in Scheme 16.

PMMA with dithiobenzoate end-groups formed by RAFT polymerization is more stable to weight loss than that formed by conventional radical polymerization (Fig. 1)[150] and thermolysis is a suitable method for cleanly removing the end-groups. Thermolysis of PMMA with dithiobenzoate end-groups was reported by Lima *et al.*,[40] Patton *et al.*,[42] Chong *et al.*[150] and Xu *et al.*[155] Thermolysis provides a macromonomer product with molecular weight and dispersity similar to those of the precursor polymer. The mechanism of weight loss of dithiobenzoate end-groups appears to be a concerted Chugaev elimination process (Scheme 18).[150] Xu *et al.*[155] reported that PMMA with dithiobenzoate end-groups degrades at substantially lower temperature than found by Chong *et al.*[150] or in other recent studies.[40,42] Factors which might lead to

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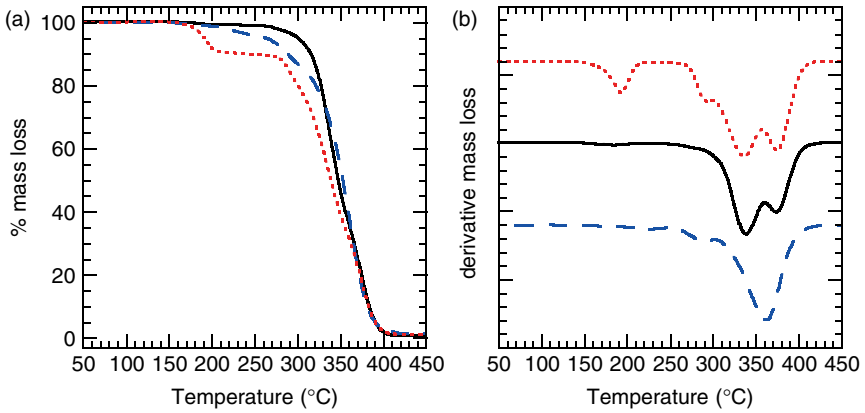
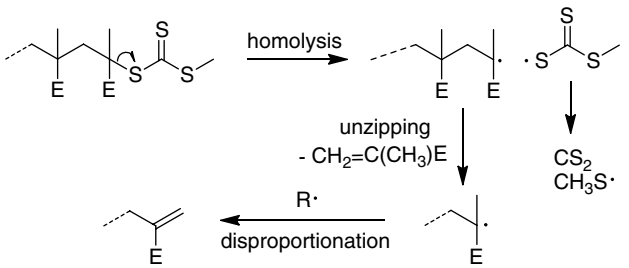
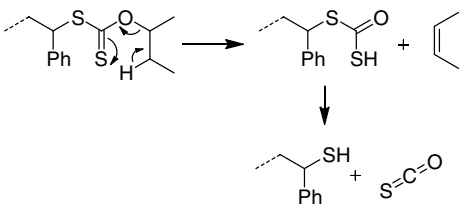


Figure 1. (a) Percentage mass loss and (b) derivative of mass loss (arbitrary units, curves are offset) for PMMA with methyl trithiocarbonate end (dotted curve), dithiobenzoate end (solid curve) and prepared by conventional free radical polymerization (dashed curve). All samples had number-average molecular weight of *ca* 50 000 g mol^{-1}. The samples were heated from 50 to 600 °C at 5 °C min^{-1} under nitrogen. (Reproduced from Chong *et al.*[150] © American Chemical Society).



Scheme 19. Thermolysis of PMMA with *S*-methyl trithiocarbonate end (E = CO_2CH_3).



Scheme 20. Thermolysis of PSt with *S*-isobutyl xanthate end.

PMMA appearing less stable include the presence of residual initiator (left over from polymer synthesis) or perhaps metal contamination in the sample. Thermolysis is less suitable for removal of the trithiocarbonate end-groups from PMMA since end-group loss is accompanied by polymer degradation and lowering of molecular weight.[150] Loss of the methyl trithiocarbonate end is believed to involve initial C–S bond homolysis to generate a PMMA propagating radical which can then degrade by unzipping (Scheme 19).

The thermolysis of xanthate-terminated polymers has also been reported.[89,156] For the case of *S*-polystyrene and *S*-poly(*t*-butyl acrylate) *O*-isobutyl xanthate at 180 °C, selective elimination to provide 2-butene and a polymer with a thiol end-group is observed (Scheme 20).[156] In contrast, for the case of *S*-poly(vinyl acetate) *O*-ethyl xanthate,[89] the products are consistent with involvement of the PVAc propagating radical and suggest a mechanism involving initial C–S bond homolysis analogous to that shown in Scheme 16.

RAFT agents in NMP

Addition–fragmentation–coupling can also be performed in the presence of a nitroxide to yield an alkoxyamine chain end.[30,157] This enables macro-RAFT agents to be used in NMP.

Favier *et al.*[30] have developed a process for exchanging the thiocarbonyl group (ZC(=S)S–) of macro-RAFT agents with a nitroxide (R′$_2$NO–) (Scheme 21). The process was termed ESARA ('exchange of substituents between (macro)alkoxyamines and (macro)RAFT agents'). The process was demonstrated for low-molecular-weight RAFT agents and macro-RAFT agents and SG1-based alkoxyamines.

RAFT agents in ATRP

Thiocarbonylthio compounds have found use as initiators in ATRP. The first reports involved the use of *N*,*N*-dialkyl dithiocarbamate derivatives as ATRP initiators.[158–161] Such reagents do not provide effective control over the polymerization of more activated monomers (MAMs), e.g. St, MMA, MA, acrylonitrile (AN), in the RAFT process so the control mechanism is most likely that of ATRP.

Recently, more active RAFT agents such as *N*,*N*-diaryldithiocarbamates (**12**),[162,163] 1-pyrrolecarbodithioates (**13**)[164] and dithioesters (1-dithionaphthalates (**14**),[165,166] dithiobenzoates[28,29]) have been used as initiators in these polymerizations. Since these are effective RAFT agents with the monomers used, it is pertinent to question the mechanism of the process. While one should not exclude ATRP participation, it seems best to consider these processes as RAFT polymerizations with ATRP initiation.

Switchable RAFT agents

RAFT agents such as dithioesters (Z = aryl or alkyl) or trithiocarbonates (Z = alkylthio), suitable for controlling polymerization of MAMs, inhibit or retard polymerizations of 'less activated' monomers (LAMs), e.g. vinyl acetate (VAc), NVP and *N*-vinylcarbazole (NVC). Similarly, RAFT agents suitable for controlling polymerizations of LAMs, such as *N*,*N*-dialkyl- or *N*-alkyl-*N*-aryl dithiocarbamates and xanthates, tend to be ineffective with MAMs.

The reduced effectiveness of dithiocarbamate and xanthate RAFT agents with MAMs relates to their lower reactivity towards radical addition and consequent lower transfer constants.[118] The double-bond character of the thiocarbonyl group is reduced

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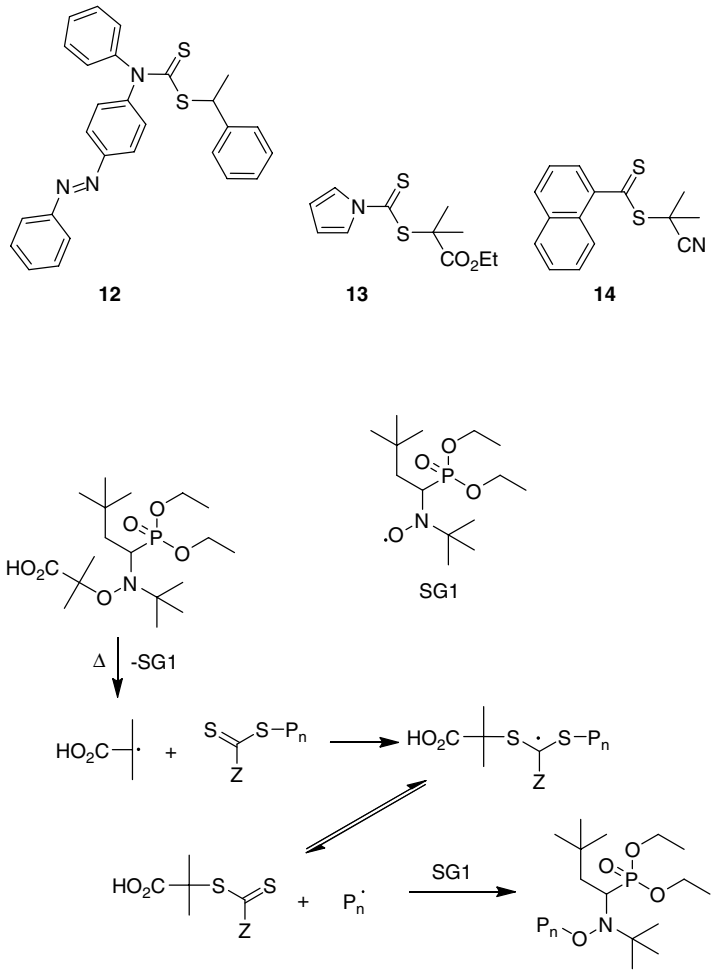
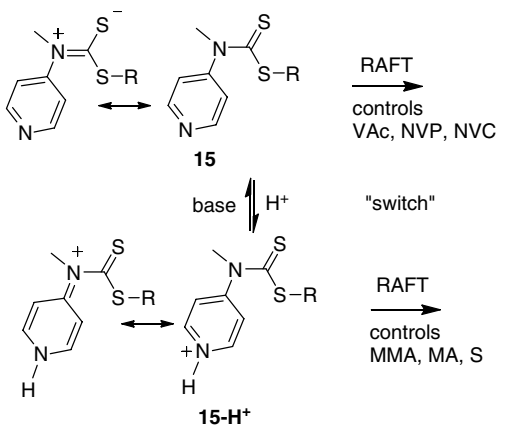


Scheme 21. Exchange of substituents between (macro)alkoxyamines and (macro)RAFT agents (ESARA).



Scheme 22. Switchable RAFT agents.

by the contribution of zwitterionic canonical forms which localize a positive charge on nitrogen and negative charge on sulfur.[118,167] On the other hand, the tendency of dithioesters or trithiocarbonates to inhibit polymerization of LAMs is a consequence of the poor radical leaving group ability, with respect to the R radical, of propagating species with a terminal LAM unit. Dithiocarbamates which possess electron-withdrawing groups adjacent to nitrogen or where the nitrogen lone pair is part of an aromatic ring system are effective with MAMs[118,167] but inhibit polymerizations of LAMs. One consequence of this is that the direct synthesis of low-dispersity polyMAM-*block*-polyLAM is difficult or not possible using the conventional range of RAFT agents.

Recently, we reported a new class of stimuli-responsive RAFT agents in which the Z group can be switched to offer good control over polymerization of both MAMs and LAMs and a route to polyMAM-*block*-polyLAM.[32] The *N*-4-pyridinyl-*N*-methyldithiocarbamates (**15**; Scheme 22) are effective in controlling the polymerization of VAc and other LAMs but give at best poor control over the polymerization of MAMs. In this, their activity is similar to *N*-phenyl-*N*-methyldithiocarbamates.[168] However, in the presence of a strong acid, the protonated form **15-H**$^+$ provides excellent control over the polymerization of MAMs. The protonated RAFT agents strongly retarded or inhibited the polymerization of LAMs. Their activity resembles that of aromatic dithiocarbamates, such as the 1-pyrrolecarbodithioates.[167]

The process is illustrated in Scheme 23 for the preparation of PMMA-*block*-PVAc. In the first step the protonated RAFT agent is used to control the polymerization of MMA to form PMMA. This macro-RAFT agent is then neutralized (or 'switched') *in situ* by adding a stoichiometric amount of *N*,*N*-dimethylaminopyridine (DMAP). RAFT polymerization of VAc then provided the desired block copolymer.

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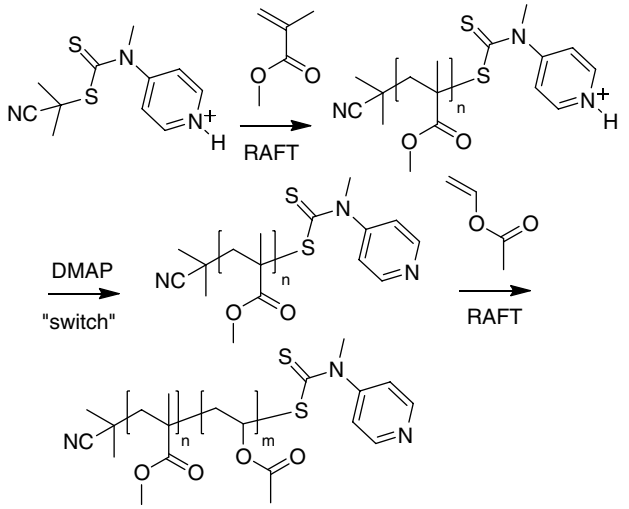


Scheme 23. Synthesis of PMMA-*block*-PVAc using switchable RAFT agent.

The R group of the RAFT agent or macro-RAFT agent still needs to be selected according to the monomer being polymerized. This has a number of consequences. In applying the method to polyMAM-*block*-polyLAM, the block comprising MAMs needed to be synthesized first.[32] This sequence is necessary because polyLAM• are poor radical leaving groups with respect to polyMAM• and, consequently, polyLAM macro-RAFT agents can have very low transfer constants in MAM polymerization. Significant retardation was observed during attempts to synthesize PSt-*block*-PVAc or other PSt-*block*-polyLAM starting from a PSt macro-RAFT agent.[31] This was attributed to the very low rate of PSt• addition to VAc. Block copolymerization of PSt and PVAc with minimal retardation was accomplished by forming a PSt-*block*-PMA as an intermediate.[31]

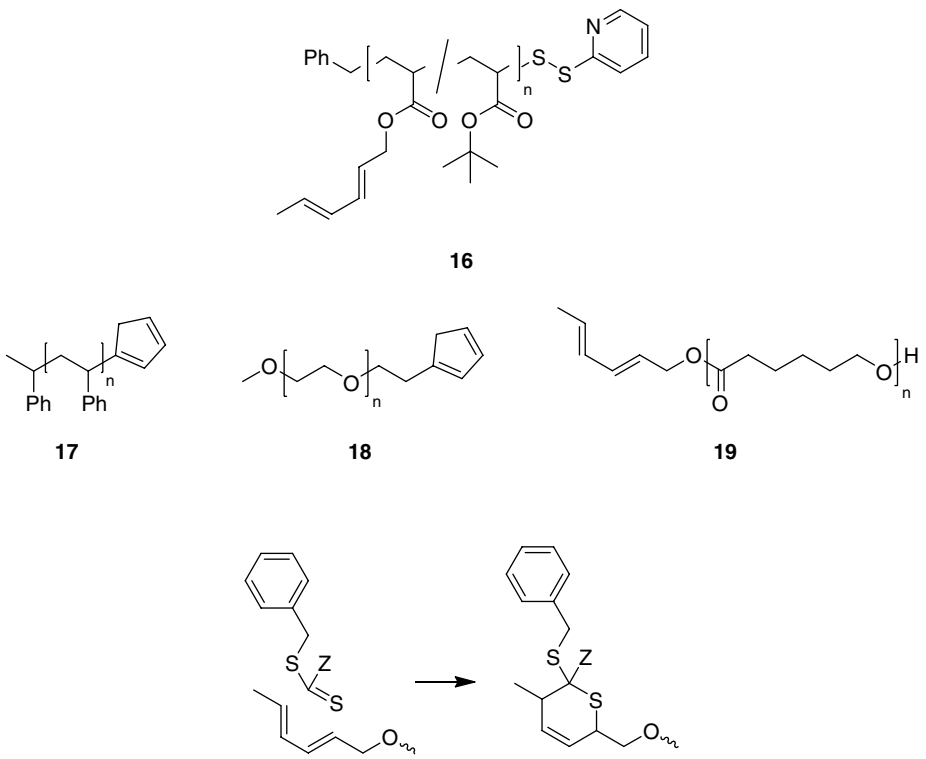
Click reactions of thiocarbonylthio compounds

Use of 'click chemistry' in polymer chemistry has recently attracted much attention.[169–172] The RAFT process can be used to synthesize polymers with clickable moieties at the chain ends through the use of RAFT agents with appropriate functionality on Z or R. A significant number of these studies concern the combination of RAFT and azide-alkyne 1,3-dipolar cycloaddition. Many RAFT agents with azido functionality or acetylene functionality have appeared.[10] In this review we only consider click reactions which directly involve the thiocarbonylthio or derived groups.

Processes that involve transformation of thiol end-groups formed from the thiocarbonylthio group by hydrolysis, aminolysis and other processes have been mentioned above and examples are included in Table 1. Many have been described as click reactions and have been used in forming biopolymer conjugates[13] and in the optoelectronic field.[19]

The [4 + 2] cycloaddition involving heterodienophiles is known as a powerful method for the synthesis of six-membered heterocycles. In particular, the thiocarbonyl group can function as a heterodienophile and provide a route to dihydrothiopyrans.[173,174] The cycloaddition is facilitated by electron-withdrawing substituents in the α-position to the thiocarbonyl and is catalysed by Lewis and protic acids.

Barner-Kowollik and co-workers took advantage of this to show that RAFT agents and macro-RAFT agents with electron-withdrawing Z groups undergo hetero-Diels–Alder reactions with suitable dienes (Scheme 24).[175–184] The process has been developed as a route to block copolymers,[176,181,185] star polymers,[175,176,179] graft copolymers[184] and modified



Scheme 24. Hetero-Diels–Alder reaction, and examples of suitable dienes.

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Table 6. Examples of end-group transformation by the hetero-Diels–Alder reaction

Terminal monomer[a]	Z[b]	Diene	Temperature (°C)	Ref.
BA, OEGA, *t*BA, St	2-Pryidyl	**16**	RT[c,d]	184
iBoA, St	2-Pryidyl	**17,18**	RT[c,d]	181
St	$EtO_2P(O)$–	**17,18**	RT[c,d]	181
St	2-Pryidyl	**19**	50[d]	176
St	$EtO_2P(O)$–	**19**	50[e]	176
iBoA	2-Pryidyl	**17**	RT[c,d]	185

[a] Monomer unit adjacent to thiocarbonylthio group.
[b] Z group of RAFT agent.
[c] RT, room temperature.
[d] Trifluoroacetic acid catalyst.
[e] Zinc chloride catalyst.
Abbreviation not defined elsewhere: OEGA, oligo(ethylene glycol) acrylate.

surfaces.[180,182] The process requires RAFT agents or macro-RAFT agents with an electron-withdrawing Z group. Suitable Z groups include pyridyl, phosphonate and phenylsulfonyl. To achieve acceptable rates, the reaction is typically catalysed by trifluoroacetic acid.[176] Examples of the process are provided in Table 6.

CONCLUSIONS

A wide variety of methods are now available for removing or transforming the thiocarbonylthio groups in RAFT-synthesized polymers. All have advantages and limitations depending on the intended application. The thiocarbonyl functionality present in RAFT-synthesized polymers, once seen as a limitation to the widespread adoption of RAFT polymerization, is now seen as an enabling functionality in addressing the needs of the biomedical, optoelectronic, nanotechnology and other sectors.

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